

Hallucination Taxonomy for RAG-Augmented Smoking Cessation Counseling

QuitTxt Protocol-Guided AI System — Supplementary Analysis

Overview

To systematically characterize errors in LLM-generated smoking cessation responses, we developed a six-category hallucination taxonomy based on analysis of 400 response pairs (100 questions × 4 configurations: Baseline, AI RAG, Human RAG, Web RAG). Each category captures a distinct failure mode observed when comparing generated answers against ground-truth references and retrieved contexts.

Taxonomy Categories

Type I: Fabricated Statistics (Factual Fabrication)

The model introduces specific numerical claims—percentages, timeframes, or statistics—that do not appear in the ground truth or retrieved context. This is the **highest-risk category** in health counseling, as fabricated statistics can mislead patients about treatment efficacy or health outcomes.

Example: When asked about secondhand smoke exposure among children, the baseline model cited specific prevalence figures and a year absent from any source material.

Prevalence: Baseline: 3%; AI RAG: 0%; Human RAG: 0%; Web RAG: 0%

Type II: Unsupported Specifics (Entity Hallucination)

The model names specific medications, products, organizations, or treatments not mentioned in the source material. While often clinically plausible, these specifics are ungrounded and may not align with evidence-based guidelines for the target population.

Example: When asked whether quitting cold turkey is effective, the baseline model recommended 'nicotine replacement therapy' despite the ground truth discussing cold turkey specifically without mentioning NRT.

Prevalence: Baseline: 4%; AI RAG: 1%; Human RAG: 0%; Web RAG: 0%

Type III: Ungrounded Authority Claims (Attribution Fabrication)

The model invokes vague appeals to research or expert consensus (e.g., 'studies have shown,' 'experts recommend') without these claims appearing in the source material. This creates a false sense of evidence-backed authority.

Example: Responses prefaced with 'research shows that...' when the source material made no such attribution.

Prevalence: Low across all configurations (<2%), but notable because it inflates perceived credibility.

Type IV: Significant Omission (Incomplete Faithfulness)

The generated answer diverges substantially from the ground truth, covering different aspects of the topic or missing critical information. Measured as <15% word overlap with the reference answer. This captures cases where the model generates topically relevant but substantively different content.

Example: When asked about teen smoking motivations, AI RAG and Web RAG responses focused on peer pressure and stress, while the ground truth emphasized advertising, flavored products, and family influence.

Prevalence: Baseline: 3%; AI RAG: 20%; Human RAG: 8%; Web RAG: 20%

Type V: Context Disconnection (Retrieval-Generation Misalignment)

In RAG configurations, the model retrieves context documents but generates an answer with minimal connection to the retrieved content (<5% context utilization). This indicates a failure in the RAG pipeline where the model defaults to parametric knowledge despite having relevant context available.

Example: For questions about erectile function and smoking, retrieved contexts discussed cardiovascular benefits, but the generated answer drew primarily on pre-trained knowledge.

Prevalence: AI RAG: 4%; Human RAG: 3%; Web RAG: 16%

Type VI: Benign Elaboration (Unsourcesd but Harmless Advice)

The model adds advisory language and practical suggestions beyond what the source material states. While not factually incorrect, this content is ungrounded and represents the boundary between helpful elaboration and unfaithful generation.

Example: Responses containing multiple advisory phrases ('you might consider,' 'it may help') not present in source material.

Prevalence: Low across all configurations (<2%).

Distribution Across Configurations

Hallucination Type	Baseline	AI RAG	Human RAG	Web RAG
I. Fabricated Statistics	3	0	0	0
II. Unsupported Specifics	4	1	0	0
III. Ungrounded Authority	<2	<2	<2	<2
IV. Significant Omission	3	20	8	20
V. Context Disconnection	—	4	3	16
VI. Benign Elaboration	<2	<2	<2	<2

Table: Hallucination type prevalence per 100 questions across experimental configurations.

Key Findings

- **RAG eliminates high-risk fabrication.** Types I and II (fabricated statistics and unsupported specifics) occur almost exclusively in the baseline configuration. All three RAG approaches effectively suppress the most dangerous hallucination categories, reducing fabricated statistics from 3% to 0%.
- **RAG introduces omission-type errors.** Type IV (significant omission) is paradoxically more prevalent in RAG configurations (8–20%) than baseline (3%). This occurs because RAG constrains the model to retrieved content, which may not cover all aspects of the ground-truth answer. The concise prompt template ('answer using ONLY facts from the knowledge base') amplifies this effect.
- **Human-curated RAG shows best overall profile.** Human RAG achieves the lowest rates across all hallucination types: zero fabrication, zero unsupported specifics, moderate omission (8%), and minimal context disconnection (3%).

- **Web RAG suffers from retrieval quality.** Web RAG shows the highest context disconnection rate (16%), indicating that its smaller corpus (278 pairs vs. 4,431–5,936) frequently retrieves tangentially related content, forcing the model to fall back on parametric knowledge.
- **Risk severity varies by type.** In a health counseling context, Types I–III pose direct patient safety risks (misinformation), while Types IV–VI represent quality-of-response issues. RAG's primary value is eliminating the former, even at the cost of increased Type IV errors.

Implications for Smoking Cessation AI Systems

This taxonomy reveals a **safety-completeness tradeoff** in RAG-augmented health counseling: stricter grounding in retrieved content reduces dangerous fabrications but increases omission of relevant information. System designers should consider hybrid approaches that use RAG for factual grounding while allowing controlled parametric elaboration for comprehensive coverage. The taxonomy also provides a framework for targeted evaluation of future smoking cessation AI systems beyond aggregate faithfulness scores.